

InternSpark internship

Wireshark packet analysis report

1 ARP Traffic Analysis

Filter Used: arp

Objective:

To analyze Address Resolution Protocol (ARP) traffic and identify communication between devices within the local network.

Observed Results:

A total of six ARP packets were captured during the monitoring session. The gateway device (192.168.31.1) transmitted ARP requests asking for the MAC address associated with IP address 192.168.31.96. The target device successfully responded with its MAC address, allowing communication between both hosts.

Packet Details:

- ARP Request: "Who has 192.168.31.96? Tell 192.168.31.1"
- ARP Reply: "192.168.31.96 is at a8:59:5f:42:85:10"

Analysis:

ARP is responsible for translating IP addresses into MAC addresses within a local area network. The captured packets confirm successful address resolution between network devices. No duplicate responses, unusual broadcasts, or conflicting MAC addresses were observed.

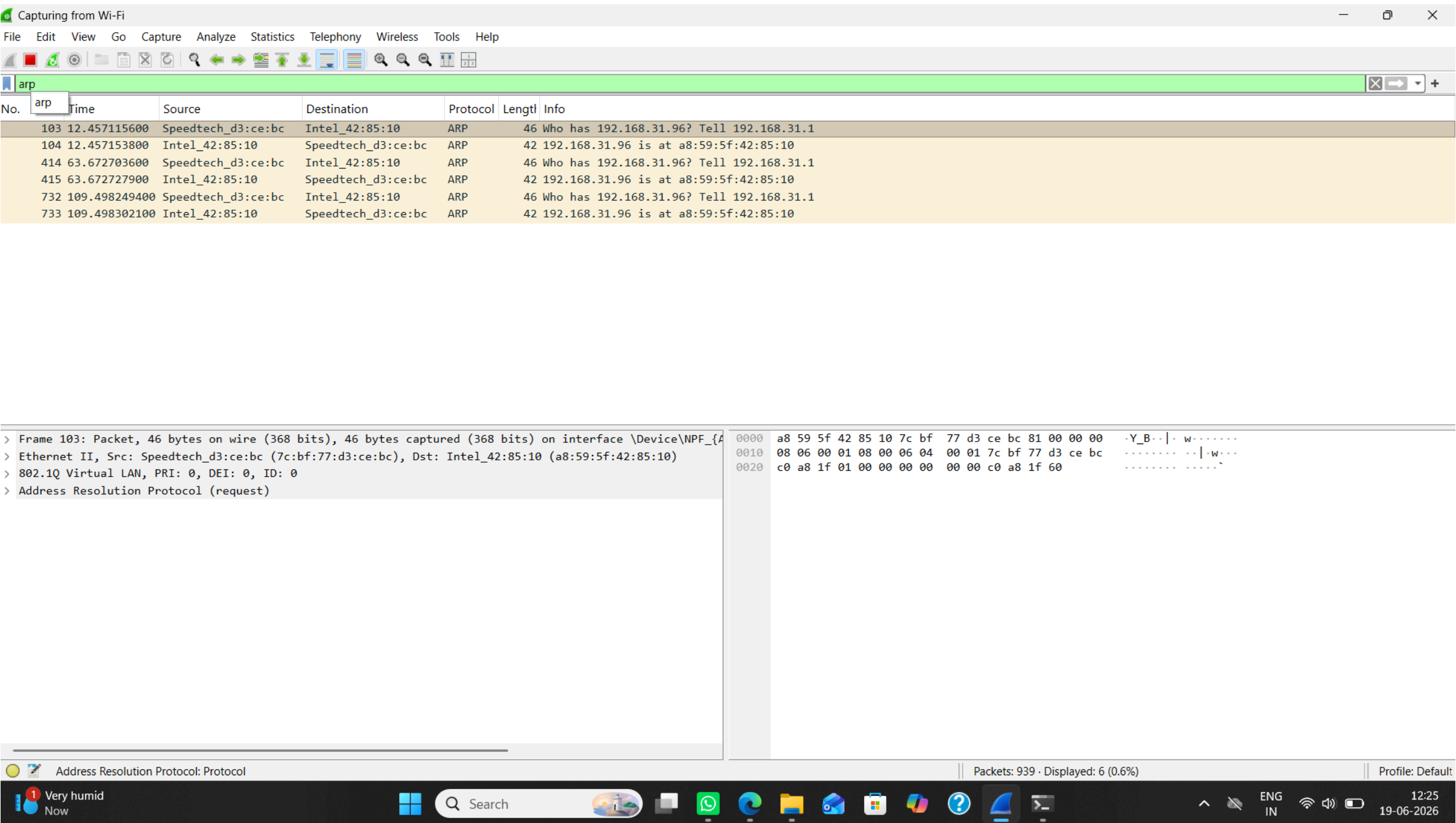
Security Assessment:

No evidence of ARP Spoofing, ARP Poisoning, Man-in-the-Middle attacks, or unauthorized MAC address advertisements was detected.

Risk Level: Informational

Conclusion:

The ARP traffic observed is normal and necessary for local network communication.



2 DNS Traffic Analysis

Filter Used: dns

Objective:
To identify domain name resolution requests and responses generated by the system.

Observed Results:
DNS requests were generated for multiple domains including:

- www.google.com
- www.bing.com
- assets.msn.com
- wpad.lan

The DNS server (192.168.31.1) successfully resolved public internet domains. The request for wpad.lan returned a "No Such Name" response.

Analysis:
DNS is an application-layer protocol that translates human-readable domain names into IP addresses. The captured traffic confirms normal browsing activity and successful communication with the configured DNS server.

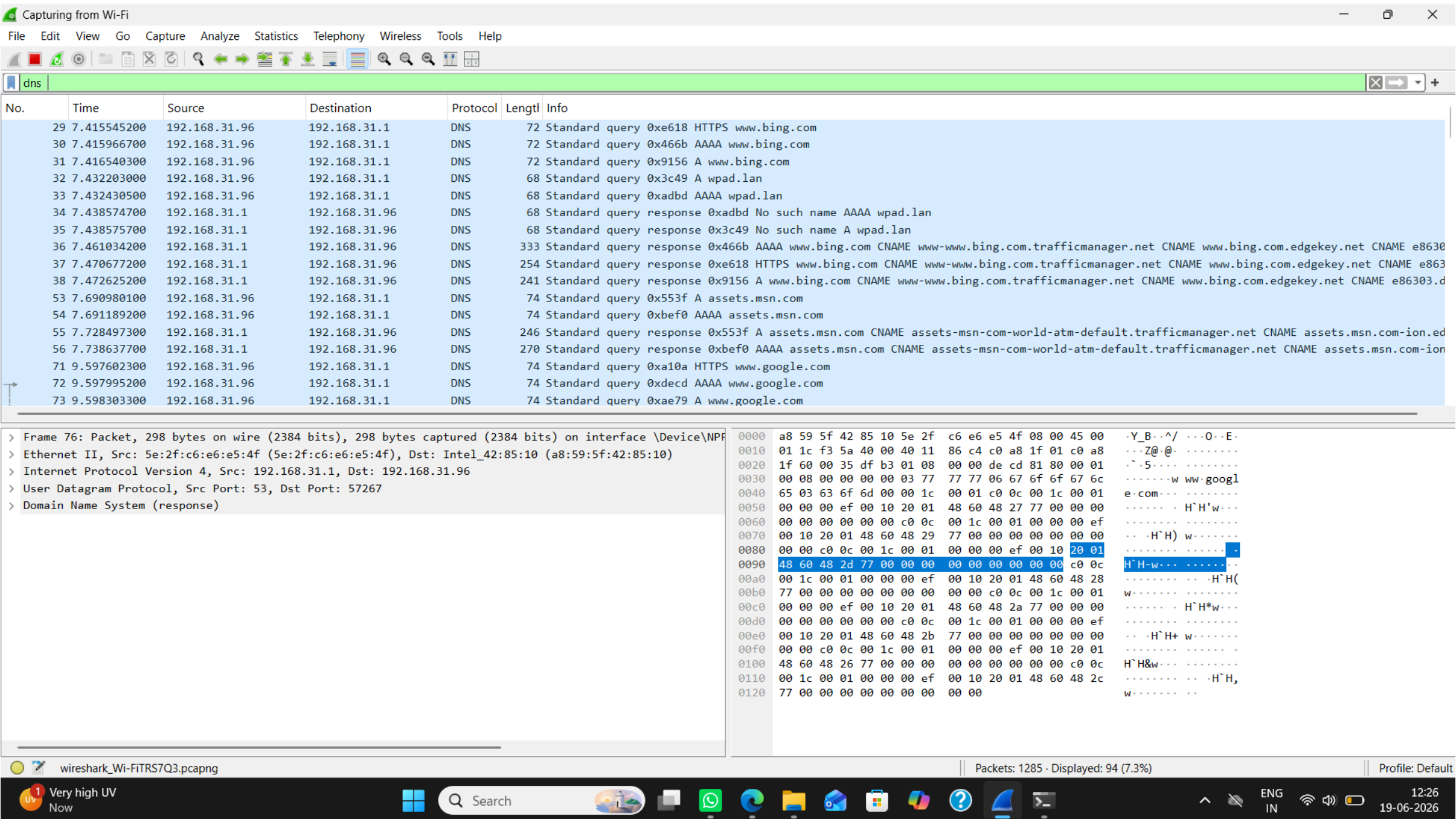
Packet Structure Analysis:

- Source IP: 192.168.31.96
- Destination IP: 192.168.31.1
- Protocol: DNS (UDP Port 53)
- Query Types: A Record, AAAA Record, HTTPS Record

Security Assessment:
No suspicious domains, DNS tunneling patterns, excessive queries, or data exfiltration attempts were identified.

Risk Level: Informational

Conclusion:
DNS operations were functioning correctly, and all observed traffic was consistent with legitimate web browsing activity.



3 TCP, TLS, and Anomaly Analysis

Filter Used: tcp, tls, tcp.analysis.flags

Objective:

To analyze TCP communication, encrypted traffic, and identify any unusual packet behavior.

Observed Results:

The packet capture displayed:

- ACK Packets
- FIN-ACK Packets
- TCP Reset (RST) Packets
- TCP Retransmissions
- Duplicate ACK Packets
- TLS 1.2 Traffic
- TLS 1.3 Traffic
- QUIC Protocol Traffic
- Client Hello Messages
- Encrypted Application Data

TCP Analysis:

TCP packets were primarily associated with HTTPS communication over Port 443. FIN and RST packets indicated normal connection termination. Retransmissions and Duplicate ACKs suggest temporary packet loss or minor network congestion.

TLS Analysis:

TLS 1.2 and TLS 1.3 protocols were observed securing communication between the client and remote servers. Client Hello and Server Hello exchanges confirmed successful session establishment. Application data remained encrypted, ensuring confidentiality and integrity.

QUIC Analysis:

QUIC traffic was detected over UDP and is commonly used by modern browsers and services such as Google, YouTube, and Microsoft applications. QUIC improves performance by reducing connection setup time.

Suspicious Packet Analysis:

The following anomalies were identified:

- TCP Retransmissions
- Duplicate ACK Packets
- TCP Reset Packets

These events are commonly observed in wireless networks and are typically caused by network congestion, packet reordering, or connection closure events.

Security Assessment:

No evidence of:

- Port Scanning
- Denial of Service (DoS)
- DNS Tunneling
- ARP Poisoning
- Malware Communication
- Unauthorized Connections

Risk Summary:

High Risk Findings: 0

Medium Risk Findings: 0

Low Risk Findings: 3

Informational Findings: 3

Recommendations:

- Continue monitoring DNS activity for suspicious domains.
- Monitor retransmission rates to identify network performance issues.
- Maintain HTTPS usage to protect sensitive communications.
- Perform periodic packet analysis to identify abnormal traffic patterns.
- Use IDS/IPS solutions for continuous network monitoring.

Final Conclusion:

The Wireshark packet capture successfully identified ARP, DNS, TCP, TLS 1.2, TLS 1.3, and QUIC protocols. Minor retransmissions and duplicate acknowledgments were observed, which are normal in wireless environments. No malicious activity, attack attempts, or security incidents were detected during the capture session. The analyzed network traffic represents normal and secure communication behavior.

Capturing from Wi-Fi

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
9	1.615644000	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
10	1.692735800	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	TLSv1.2	144	Application Data
11	1.745250300	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	TCP	74	443 → 34124 [ACK] Seq=1 Ack=71 Win=327 Len=0
12	2.023818100	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	TLSv1.2	146	Application Data
13	2.067174400	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	TCP	74	34124 → 443 [ACK] Seq=71 Ack=73 Win=252 Len=0
22	5.508388100	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TCP	98	[TCP Retransmission] 443 → 47085 [FIN, PSH, ACK] Seq=1 Ack=1 Win=504 Len=24
59	8.119033000	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	TCP	75	19611 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
61	8.160598600	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	TCP	74	443 → 19611 [ACK] Seq=1 Ack=2 Win=325 Len=0
83	10.345394500	192.168.31.96	104.208.16.89	TCP	54	2741 → 443 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
94	12.106105700	2409:40f2:1181:337e::...	2404:6800:4003:c00::...	TCP	74	46342 → 5228 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
102	12.455006400	2409:40f2:1181:337e::...	2404:6800:4003:c00::...	TCP	74	[TCP Retransmission] 46342 → 5228 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
105	12.540999400	2404:6800:4003:c00::...	2409:40f2:1181:337e::...	TCP	74	5228 → 46342 [RST] Seq=1 Win=0 Len=0
106	13.093726400	192.168.31.96	40.79.163.155	TCP	55	3020 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
107	13.288926600	40.79.163.155	192.168.31.96	TCP	66	443 → 3020 [ACK] Seq=1 Ack=2 Win=16386 Len=0 SLE=1 SRE=2
108	13.699039600	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TCP	98	[TCP Retransmission] 443 → 47085 [FIN, PSH, ACK] Seq=1 Ack=1 Win=504 Len=24
109	14.107977300	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
110	14.107979600	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TCP	74	443 → 35485 [FIN, ACK] Seq=25 Ack=1 Win=564 Len=0

> Frame 61: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{A2...}

> Ethernet II, Src: Speedtech_d3:ce:bc (7c:bf:77:d3:ce:bc), Dst: Intel_42:85:10 (a8:59:5f:42:85:10)

> Internet Protocol Version 6, Src: 2a03:2880:f354:120:face:b00c:0:167, Dst: 2409:40f2:1181:337e:102:85ae:6b43

> Transmission Control Protocol, Src Port: 443, Dst Port: 19611, Seq: 1, Ack: 2, Len: 0

0000 a8 59 5f 42 85 10 7c bf 77 d3 ce bc 86 dd 60 00 .Y_B...|:w.....

0010 00 00 00 14 06 36 2a 03 28 80 f3 54 01 20 fa ce6*...T...

0020 b0 0c 00 00 01 67 24 09 40 f2 11 81 33 7e 01 02g\$.@...3~...

0030 85 ae 6b 43 87 87 01 bb 4c 9b 4f 9b f1 26 8f 30 ..kC....L-O...&-0

0040 36 21 50 10 01 45 43 75 00 00 00 00 00 00 00 6!P...ECu...

Transmission Control Protocol: ProtocolPackets: 1567 · Displayed: 526 (33.6%)Profile: Default

Capturing from Wi-Fi

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tls

No.	Time	Source	Destination	Protocol	Length	Info
9	1.615644000	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
10	1.692735800	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	TLSv1.2	144	Application Data
12	2.023818100	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	TLSv1.2	146	Application Data
39	7.475234400	2409:40f2:1181:337e::...	2405:200:1608:1731::...	QUIC	1292	Initial, DCID=b0da3d6749288670, PKN: 1, PING, CRYPTO, PING, PADDING, PING, PING, CRYPTO, PING, PING, PADDING, CRYPTO, PADDING, PING,
40	7.475531500	2409:40f2:1181:337e::...	2405:200:1608:1731::...	QUIC	1292	Initial, DCID=b0da3d6749288670, PKN: 2, PADDING, PING, PADDING, PING, PING, PADDING, CRYPTO, PING, CRYPTO, PADDING, CRYPTO, PING, PAD
41	7.532477400	2405:200:1608:1731::...	2409:40f2:1181:337e::...	QUIC	1292	Initial, SCID=0a42db479d806d24, PKN: 3, CRYPTO, PADDING
109	14.107977300	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
119	15.950515000	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
127	16.565111400	2405:200:1608:1731::...	2409:40f2:1181:337e::...	TLSv1.2	98	Application Data
141	19.995334400	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	TLSv1.2	144	Application Data
146	20.290668100	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	TLSv1.2	146	Application Data
151	20.351353800	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	QUIC	1292	Initial, DCID=63e09ddd2826edfb, PKN: 1, PING, CRYPTO, PING, CRYPTO, CRYPTO, CRYPTO, CRYPTO, PING, CRYPTO, PADDING, PING, PADDING, CRY
152	20.351463900	2409:40f2:1181:337e::...	2a03:2880:f354:120::...	QUIC	1292	Initial, DCID=63e09ddd2826edfb, PKN: 2, PING, CRYPTO, PADDING, PING, PADDING, PING, PING, PING, CRYPTO, PADDING, PING, PADDING, PING,
155	20.415721000	2a03:2880:f354:120::...	2409:40f2:1181:337e::...	QUIC	1294	Initial, SCID=bb004d4500cbad0a, PKN: 1970004, CRYPTO, ACK, PADDING
203	27.031592100	192.168.31.96	23.193.56.33	TLSv1.3	524	Client Hello (SNI=catalog.gamepass.com)
205	27.094099000	23.193.56.33	192.168.31.96	TLSv1.3	1316	Server Hello, Change Cipher Spec, Application Data
211	27.150645600	23.193.56.33	192.168.31.96	TLSv1.3	1232	Application Data, Application Data

> Frame 41: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{A2...}

> Ethernet II, Src: Speedtech_d3:ce:bc (7c:bf:77:d3:ce:bc), Dst: Intel_42:85:10 (a8:59:5f:42:85:10)

> Internet Protocol Version 6, Src: 2405:200:1608:1731::312c:71b0, Dst: 2409:40f2:1181:337e:102:85ae:6b43

> User Datagram Protocol, Src Port: 443, Dst Port: 51438

> QUIC IETF

0000 a8 59 5f 42 85 10 7c bf 77 d3 ce bc 86 dd 60 0a .Y_B...|:w.....

0010 14 f1 04 d6 11 3c 24 05 02 00 16 08 17 31 00 00<\$.....1..

0020 00 00 31 2c 71 b0 24 09 40 f2 11 81 33 7e 01 02 ..1,q\$.@...3~...

0030 85 ae 6b 43 87 87 01 bb c8 ee 04 d6 ee 9c c8 00 ..kC....L-O...&-0

0040 00 00 01 00 08 0a 42 db 47 9d 80 6d 24 00 44 bcB...G...m\$.D-

0050 7e f3 52 7d 6f 28 cd b4 5e 9a 95 86 b5 8a 95 5c ~R)o(-...^.....\

0060 82 a2 6c 6c d3 0b ae 6b 65 fb 56 df 07 87 e5 5c ...1l...k e-V....\

0070 e1 df a0 2a f8 d6 f0 6d 05 63 0a d9 8d b2 0b 9d ...m...c.....

0080 8b 31 2b f4 9a bc e5 bf ba 80 52 46 51 c0 50 29 ..1+.....RFQ-P)

0090 d1 00 3d 01 1c 9d 8a 47 f5 4d 80 79 1f 9a 1f 08 ==+.....G -M-y....

00a0 ed 8a b6 5c 24 f1 8b a2 d0 b4 9b 1f 56 41 4c 70 ...\\$.....VALp...

00b0 5b ca 0b 82 20 cb bc 41 14 22 41 3e ad 82 3c dc [...-A -"A>-<...

00c0 a8 b7 08 a4 9a 7f ec e0 36 5b f5 c4 0e 28 cd 786[...(-x...

00d0 61 9a 77 34 ff 99 65 52 4b 6e 87 fe 8e 2b 91 02 a-w4...eR Kn...+...

00e0 1a 5a ba 9e 99 50 19 e6 01 ca be 0f fd 5f 2c 13 -Z...P...

00f0 92 68 0b 09 cb 60 e3 dc 27 fe f2 0e c1 df fb f1 -h...

0100 1e b0 a3 88 1c 24 9d d8 d5 35 58 ef f2 2c 58 28 ...\$....5X...X(

0110 a7 25 7d e8 10 3b d8 17 d1 45 f6 de 23 66 66 90 .%);...E-#ff...

0120 5b 26 92 1f 37 ae c5 a0 c8 2e d9 58 61 f1 66 ec [&-7... ..Xa-f...

0130 57 ab 85 9a 12 04 a0 c2 fc aa 76 43 10 fb 42 b6 W.....vC...B-

0140 85 b6 dc 4b 98 c4 7c ae 1a 5f 54 86 4b eb e6 d4 ...K...|:.._T-K...

Packet (1292 bytes)Decrypted QUIC (1195 bytes)Packets: 1907 · Displayed: 287 (15.0%)Profile: Default

